4R/2025 MAT4304C

2025

MATHEMATICS

(Major–III)

Paper : MAT4304C

(Analytical Geometry)

Full Marks : 60

Time : 2½ hours

The figures in the margin indicate full marks for the questions.

1. Answer the following : $1\times7=7$

(a) Find the transformed equation of $\frac{x}{a}+\frac{y}{b}=2$ when the origin is shifted to (a,b).

(b) Write the condition so that the equation $ax^2+2hxy+by^2+2gx+2fy+c=0$ represents a parabola.

2026/5/2 13:56

(c) Write the condition so that the conic $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ has infinitely many centres.

(d) Find the centre and the radius of the sphere given by the equation

$x^2 + y^2 + z^2 - 2x - 4y + 6z = 22$.

(e) If the vertex is the origin and the Z–axis is the axis of the cone, then write the equation of the cone.

(f) Write the equation of the chord to the parabola $y^2 = 4ax$ at (x_1, y_1).

(g) Write the equation of the hyperboloid of two sheets.

2. Answer the following : $2 \times 4 = 8$

(a) Find the value of K, so that the equation

$$Kxy - 8x + 9y - 12 = 0$$

represents a pair of straight lines.

(b) Find the angle between the lines represented by the equation

$6x^2 + 13xy + 6y^2 + 8x + 7y + 2 = 0$.

(c) Find the equation of the sphere through the circle $x^2+y^2+z^2=9$, $2x+3y+4z=5$ and the origin.

(d) Find the equation of the cone whose vertex is at the origin and the guiding curve is given by

$$x=a,\ y^2+z^2=b^2.$$

3. Answer the following: $5\times3=15$

(a) Show that the equation

$$ax^2+2hxy+by^2+2gx+2fy+c=0$$

represents a a pair of parallel straight lines if

$$\frac{a}{h}=\frac{h}{b}=\frac{g}{f}.$$

(b) Transform the equation

$$11x^2-4xy+14y^2-58x-44y+126=0$$

into the new axis given by the equations are

$$x-2y+1=0,\quad 2x+y-8=0.$$

Or

(c) Show that the pair of lines

$ax^2 + 2hxy + by^2 = 0$ and

$a'x^2 + 2h'xy + b'y^2 = 0$ have one line in common if

$$4(ah' - a'h)(hb' - h'b) = (ab' - a'b)^2.$$

(d) A sphere of radius r passes through the origin O and meets at axes at A, B, C. Prove that the locus of the foot of the perpendicular to the plane ABC from O is given by

$$(x^2 + y^2 + z^2)^2 (x^{-2} + y^{-2} + z^{-2}) = 4r^2$$

Or

(e) Find the laws of the point of intersection of three mutually perpendicular tangent planes to a central conicoid.

4. Answer **either** (a) **or** (b) : 10

(a) (i) Reduce the equation

$$x^2 + 4xy + y^2 - 2x + 2y + 6 = 0$$

to the standard form. 5

2026/5/2 13:58

(ii) Find the pole of the line $x+y-1=0$ with respect to the curve 5

$$3x^2+4xy-2y^2-5x+7y-10=0$$

(b) (i) Find the equation of the chord of the conic

$\frac{l}{r}=1+e\cos\theta$ joining the two points on the conic whose vectorial angles are $(\alpha+\beta)$ and $(\alpha-\beta)$. 5

(ii) Find the equation of the pair of tangents from a point (x_1, y_1) to the conic 5

$$ax^2+2hxy+by^2+2gx+2fy+c=0$$

5. Answer **either** (a) **or** (b) : 10

(a) (i) Derive the condition that the general second degree homogeneous equation represents a right circular cone. 5

(ii) Find the equation of the right circular cylinder whose axis is $x=2y=-z$ and radius is 4. 5

2026/5/2 13:58

(b) (i) Find the equation of the tangent plane at a point (α, β, γ) to the conicoid $ax^2 + by^2 + cz^2 = 1$ 5

(ii) Prove that the locus of points from which three mutually perpendicular planes can be drawn to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1,\ z = 0$ is the sphere

$x^2 + y^2 + z^2 = a^2 + b^2$. 5

6. Answer **either** (a) **or** (b) : 10

(a) (i) Evaluate

$\int_C \vec{F}.d\vec{r}$ where $\vec{F} = (x^2 - y^2)\hat{i} + xy\hat{j}$ and C is the arc of the curve $y = x^3$ from (0,0) to (2,8). 5

(ii) Evaluate $\iint_S \vec{F}.\hat{n}\, ds$ where $\vec{F} = Z\hat{i} + x\hat{j} - 3y^2 z\hat{k}$ and S is the surface of the cylinder $x^2 + y^2 = 16$ included in the first octant between $z = 0$ and $z = 5$. 5

(b) (i) Evaluate $\int_C \vec{F}.d\vec{r} = (x^2 + y^2)\hat{i} - 2xy\hat{j}$ where C is the rectangle in the xy-plane bounded by $y = 0$, $x = a$, $y = b$, $x = 0$. 5

(ii) Evaluate $\iiint_V \phi dV$, where $\phi = 45x^2y$ and V is the closed region bounded by the planes $4x + 2y + z = 8, z = 0, x = 0, y = 0$. 5

$l_1 x + m_1 x + n_1$

$= \sqrt{l_1^2 a^2 + m_1^2 b^2}$

$\frac{x\,dx}{\sqrt{16-x^2}}$

$-\frac{dt}{t}$

$= -\ln t$

$= -\ln (16-x^2 = t^2$

$-2x\,dx = 2t\,dt$

$-x\,dx = dt$

$= (-\ln\sqrt{16-x^2})$

$\frac{1}{2} t^{-1} dt$

$2^5 = 32$

2×2

$64 \times 2 = 128$